INCH-POUND
MS35690G
22 April 2022
Superseding
MS35690F
8 April 2005

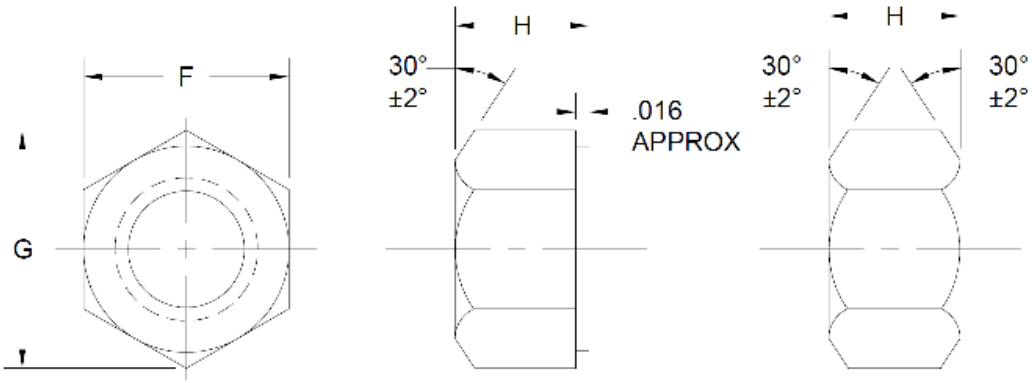
DETAIL SPECIFICATION SHEET

NUT, PLAIN, HEXAGON, UNC-2B AND UNF-2B

Inactive for new design after 22 April 2022.
For new design, use nuts as specified per ASME B18.24.

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and FF-N-836.



OPTIONAL DESIGNS

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or by sent email to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5310

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TABLE I. Design configurations.

NOMINAL SIZE OR BASIC MAJOR DIAMETER OF THREAD	WIDTH ACROSS FLATS F		WIDTH ACROSS CORNERS G		THICKNESS H	
	MAX	MIN	MAX	MIN	MAX	MIN
1/4 - 0.2500	0.4375	0.428	0.505	0.488	0.226	0.212
5/16 - 0.3125	0.5000	0.489	0.577	0.557	0.273	0.258
3/8 - 0.3750	0.5625	0.551	0.650	0.628	0.337	0.320
7/16 - 0.4375	0.6875	0.675	0.794	0.768	0.385	0.365
1/2 - 0.5000	0.7500	0.736	0.866	0.840	0.448	0.427
9/16 - 0.5625	0.8750	0.861	1.010	0.982	0.496	0.473
5/8 - 0.6250	0.9375	0.922	1.083	1.051	0.559	0.535
3/4 - 0.7500	1.1250	1.088	1.299	1.240	0.665	0.617
7/8 - 0.8750	1.3125	1.269	1.516	1.447	0.776	0.724
1 - 1.0000	1.5000	1.450	1.732	1.653	0.887	0.831
1-1/8 - 1.1250	1.6875	1.631	1.949	1.859	0.999	0.939
1-1/4 - 1.2500	1.8750	1.812	2.165	2.066	1.094	1.030
1-3/8 - 1.3750	2.0625	1.994	2.382	2.273	1.206	1.138
1-1/2 - 1.5000	2.2500	2.175	2.598	2.480	1.317	1.245
1-3/4 - 1.7500	2.6250	2.538	3.031	2.893	1.540	1.460
2 - 2.0000	3.0000	2.900	3.464	3.306	1.763	1.675
2-1/4 - 2.2500	3.3750	3.262	3.897	3.719	1.970	1.874
2-1/2 - 2.5000	3.7500	3.625	4.330	4.133	2.193	2.089
2-3/4 - 2.7500	4.1250	3.988	4.763	4.546	2.415	2.303
3 - 3.0000	4.5000	4.350	5.196	4.959	2.638	2.518

REQUIREMENTS:

1. Material: Steel, carbon, in accordance with procurement specification.
   Steel, corrosion resistant, types 303 (UNS S30300), 304 (UNS S30400), 305 (UNS S30500), 410 (UNS S41000), 416 (UNS S41600), 430 (UNS S43000) in accordance with ASTM A276/A276M and ASTM A582/A582M.
   Brass, alloy 462 (UNS C46200) hard or 464 (UNS C46400) half-hard in accordance with ASTM B21/B21.

2. Protective coating and surface treatment:

   Cadmium plated in accordance with procurement specification [1/]
   Zinc coating, ASTM B633, Fe/Zn5 (5 µm thick), Type II [2/]
   Phosphate coating, MIL-DTL-16232, Type Z, Class 2.
   Tin plated, 0.0001 thick in accordance with ASTM B545, Class A.
   Cleaned, descaled and passivated, in accordance with ASTM A380/A380M (see Table IV).
   Nickel plated, in accordance with procurement specification (see Table II).

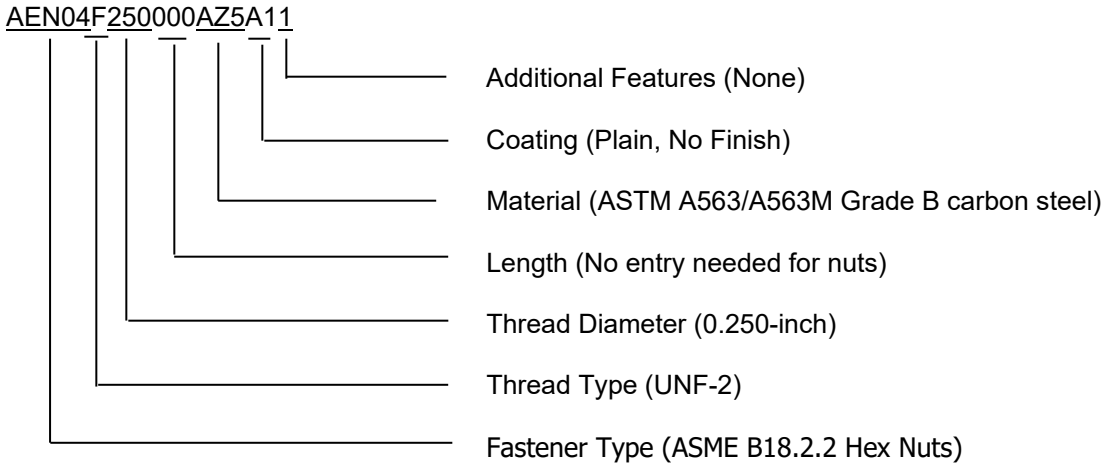
[1/] Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K16 is the preferred replacement plating. The ASME B18.24 finish code is QX.
[2/] Hexavalent Chromium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K20 is the preferred replacement plating. The ASME B18.24 finish code is QW.

3. Threads: Threads shall be in accordance with FED-STD-H28/2. Acceptability of screw threads shall be in accordance with FED-STD-H28/20, System 21.

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4. Part or identifying number (PIN): The PIN shall be constructed using ASME B18.24 (see Table II).

Example PIN code from ASME B18.24 (equivalent to MS35690-424):



MS35690-424 indicates – Nut, Plain, Hexagon, Thread size 0.2500-28 UNF-2B; Steel, Carbon, Grade B, Uncoated.

5. Dimensions in inches.

6. Referenced documents shall be of the issue in effect on the date of invitation for bids or request for proposal, except that referenced adopted documents shall give the date of the issue adopted.

7. For design feature proposes, this standard takes precedence over procurement documents referenced herein.

8. Class I and II ozone depleting substances (ODS) shall not be used in MS35690 or any referenced procedures.

9. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

ACTIVE FOR NEW DESIGN

The following Part Identifying Numbers (PIN's) are active for new design after 22 April 2022.

The information to specify the nut size, material, grade, and finish are found in ASME B18.24 (2020). The following information is provided to assist in building an ASME PIN code:

"AEN04" is the prefix for ASME B18.2.2 hex nuts
"AZ5" is the material code for Grade B nuts per ASTM A563/A563M
"AZ9" is the material code for Grade C nuts per ASTM A563/A563M
There are multiple codes in Table 5-1 for stainless steel and non-ferrous material options
"QW" or "QX" are the preferred finish codes for Zinc Nickel Electroplating of carbon and alloy steel nuts (see section 2 footnotes)

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Table II. Dash Numbers & Replacement ASME B18.24 PIN Code.

Thread Size	0.250-28 UNF-2B		0.375-16 UNC-2B		0.375-24 UNF-2B	
Material	Dash No.	ASME PIN (for reference)	Dash No.	ASME PIN (for reference)	Dash No.	ASME PIN (for reference)
Steel, Carbon, Grade B, Uncoated	424	AEN04F250000AZ5A11	604	AEN0C375000AZ5A11	624	AEN04F375000AZ5A11
Steel, Carbon, Grade C, Uncoated	427	AEN04F250000AZ9A11	607	AEN04C375000AZ9A11	627	AEN04 F375000AZ9A11
Brass, Nickel Plated	413	AEN04F250000BE4A11 Or AEN04F250000BE6A11	613	AEN04C375000BE4A11 Or AEN04C375000BE6A11	633	AEN04F375000BE4A11 Or AEN04F375000BE6A11

INACTIVE FOR NEW DESIGN

The following part numbers remain inactive for design after 30 September 1965. Use for maintenance of existing equipment only.

Table III. Inactive part numbers.

Thread Size	0.2500-20UNC-2B		0.3125-18UNC-2B	0.3125-24UNF-2B
Material	Dash No.		Dash No.	Dash No.
Steel, Carbon, Grade B,Uncoated Steel, Carbon, Grade C,Uncoated	404 407		504 507	524 527
Thread Size	0.4375-14UNC-2B	0.4375-20UNF-2B	0.5000-13UNC-2B	0.5000-20UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B,Uncoated Steel, Carbon, Grade C,Uncoated	704 707	724 727	804 807	824 827
Thread Size	0.5625-12UNC-2B	0.5625-18UNF-2B	0.6250-11UNC-2B	0.6250-18UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B,Uncoated Steel, Carbon, Grade C,Uncoated	904 907	924 927	1004 1007	1024 1027
Thread Size	0.7500-10UNC-2B	0.7500-16UNF-2B	0.8750-9UNC-2B	0.8750-14UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B,Uncoated Steel, Carbon, Grade C,Uncoated	1204 1207	1224 1227	1404 1407	1424 1427
Thread Size	1.0000-8UNC-2B	1.0000-12UNF-2B	1.1250-7UNC-2B	1.1250-12UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B,Uncoated Steel, Carbon, Grade C,Uncoated	1604 1607	1624 1627	1804 1807	1824 1827
Thread Size	1.2500-7UNC-2B	1.2500-12UNF-2B	1.3750-6UNC-2B	1.3750-12UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B,Uncoated Steel, Carbon, Grade C,Uncoated	2004 2007	2024 2027	2204 2207	2224 2227
Thread Size	1.5000-6UNC-2B	1.5000-12UNF-2B	1.7500-5UNC-2B	2.0000-4.5UNC-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B,Uncoated Steel, Carbon, Grade C,Uncoated	2404 2704	2424 2427	2804 2807	3204 3207
Thread Size	2.2500-4.5UNC-2B	2.5000-4UNC-2B	2.7500-4UNC-2B	3.0000-4UNC-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B,Uncoated Steel, Carbon, Grade C,Uncoated	3604 3607	4004 4007	4404 4407	4804 4807

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CANCELLED PART NUMBERS

The following parts are cancelled for design after 30 September 1965. For interchangeability see Tables V and VI of this standard and MS51967 thru MS51972.

Table IV. Cancelled part numbers.

Thread Size	0.2500-20UNC-2B	0.2500-28UNF-2B	0.3125-18UNC-2B	0.3125-24UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 401 402 403	 421 422 423	 501 502 503	 521 522 523
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 405 406	 425 426	 505 506	 525 526
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 408 409	 428 429	 508 509	 528 529
Steel, CRES Passivated	410	430	510	530
Brass Uncoated Tin plated	411 412	431 432	511 512	531 532
Thread Size	0.3750-16UNC-2B	0.3750-24UNF-2B	0.4375-14UNC-2B	0.4375-20UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 601 602 603	 621 622 623	 701 702 703	 721 722 723
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 605 606	 625 626	 705 706	 725 726
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 608 609	 628 629	 708 709	 728 729
Steel, CRES Passivated	610	630	710	730
Brass Uncoated Tin plated	611 612	631 632	711 ---	731 ---
Thread Size	0.5000-13UNC-2B	0.5000-20UNF-2B	0.5625-12UNC-2B	0.5625-18UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 801 802 803	 821 822 823	 901 902 903	 921 922 923
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 805 806	 825 826	 905 906	 925 926
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 808 809	 828 829	 908 909	 928 929
Steel, CRES Passivated	810	830	910	930
Brass, Uncoated	811	831	911	931

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Table IV. Cancelled part numbers. – Continued.

Thread Size	0.6250-11UNC-2B	0.6250-18UNF-2B	0.7500-10UNC-2B	0.7500-16UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 1001 1002 1003	 1021 1022 1023	 1201 1202 1203	 1221 1222 1223
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 1005 1006	 1025 1026	 1205 1206	 1225 1226
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 1008 1009	 1028 1029	 1208 1209	 1228 1229
Steel, CRES Passivated	1010	1030	1210	1230
Brass, Uncoated	1011	1031	1211	1231
Thread Size	0.8750-9UNC-2B	0.8750-14UNF-2B	1.0000-8UNC-2B	1.0000-12UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 1401 1402 1403	 1421 1422 1423	 1601 1602 1603	 1621 1622 1623
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 1405 1406	 1425 1426	 1605 1606	 1625 1626
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 1408 1409	 1428 1429	 1608 1609	 1628 1629
Steel, CRES Passivated	1410	1430	1610	1630
Brass, Uncoated	1411	1431	1611	1631
Thread Size	1.1250-7UNC-2B	1.1250-12UNF-2B	1.2500-7UNC-2B	1.2500-12UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 1801 1802 1803	 1821 1822 1823	 2001 2002 2003	 2021 2022 2023
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 1805 1806	 1825 1826	 2005 2006	 2025 2026
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 1808 1809	 1828 1829	 2008 2009	 2028 2029
Steel, CRES Passivated	1810	1830	2010	2030
Brass, Uncoated	1811	1831	2011	2031

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Table V. Interchangeability table of cancelled and superseding part numbers.

Part Numbers		Part Numbers		Part Numbers	
Cancelled	Superseded by	Cancelled	Superseded by	Cancelled	Superseded by
MS35690-401	MS35690-404	MS35690-1001	MS35690-1004	MS35690-2021	MS35690-2024
501	504	1021	1024	2201	2204
521	524	1201	1204	2221	2224
701	704	1221	1224	2401	2404
721	724	1401	1404	2421	2424
801	804	1421	1424	2801	2804
821	824	1601	1604	3201	3204
901	904	1621	1624	3601	3604
921	924	1801	1804	4001	4004
		1821	1824	4401	4404
		2001	2004	4801	4804

The following cancelled nuts are superseded by the reinstated nuts indicated below.

Table VI. Interchangeability table of cancelled and superseding part numbers.

Part Numbers		Part Numbers		Part Numbers	
Cancelled	Superseded by	Cancelled	Superseded by	Cancelled	Superseded by
MS35690-421	MS35690-424 427	MS35690-601	MS35690-604 607	MS35690-621	MS35690-624 627

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – SH
Air Force – 99
DLA – GS

Preparing activity:
Army – AT

(Project 5310-2022-013)

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